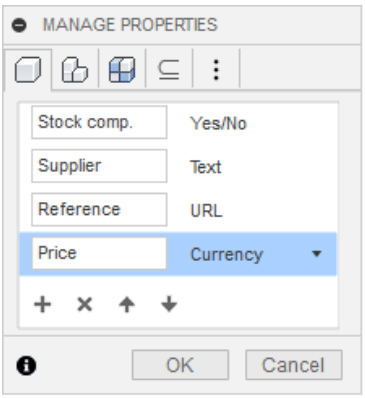
Properties ++

for Fusion 360

1 Set your custom properties

Pg 3

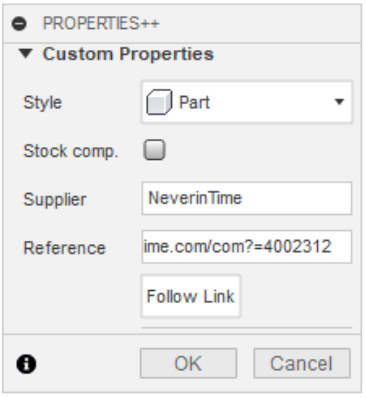
Go to *MODIFY*> *Manage Properties*. Set up the desired custom properties for each property style (*part*, *assembly* or *design*), Or re-use previously set up properties using a Template or a Standard.



2 Add data to components

Pg 6

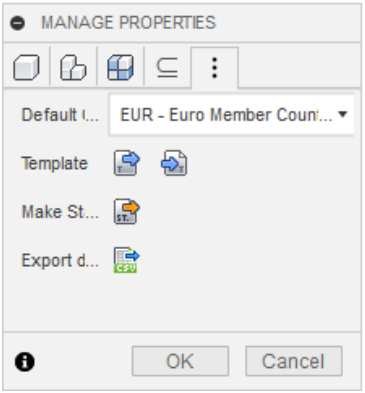
In the (design) browser; right click on the component that you would like to setup and select *Properties ++*. Add all data that you would like to associate with this component.



3 Export to table

Pg 7

Go back to *MODIFY*> *Manage Properties*. The last tab of the menu will show the options to export your data as a table which can be opened by your favorite spreadsheet software.



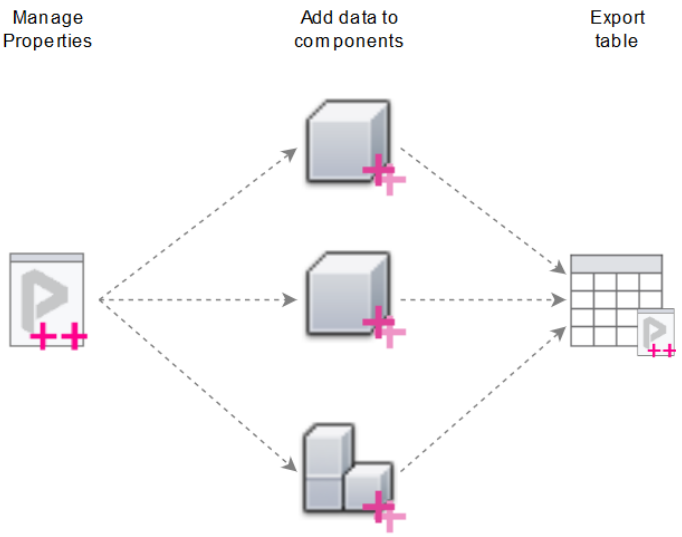
Properties ++ is an add-in for Fusion 360 which aims to help you at managing your data. This data management is based on adding custom properties to your components and designs. When your design is done you could easily export your custom data as if it were part of a Bill of Materials.

Properties are not set per component but per Component Style. A component style is a group of properties that can be reused for different components. There is a fixed number of styles; properties for parts, properties for assemblies and properties for designs (generally the root component). It is also possible to add properties to all of these styles. Setting these properties will be done using the Property Manager. Note that these settings can be exported for a different design.

For each component you can select the style that fits the components purpose. (Although in most cases the style fits the type of component and it will not need to be changed.) When the correct style is selected you will see the custom properties that belong to this component. Now you can fill in the belonging data. This data will be called the Custom Properties Data.

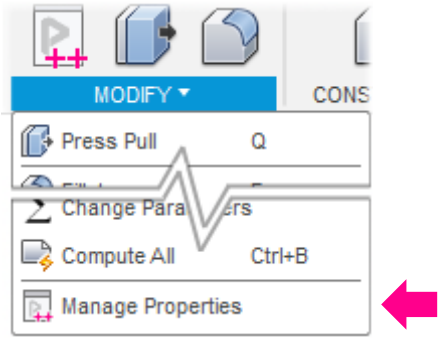
Once all data has been added to your design you could export all your data as a table. The table will be formatted as a Bill of Materials. The export file-format will be .csv so you will be able to open this table with any major spreadsheet editing tool.

This document gives further details on how to use Properties++.



1. Set your custom properties

When a new document is opened or Properties++ has just been installed there are no custom properties available just yet. The first thing we'll have to do is set these up. Open the Fusion 360 *MODIFY* panel and select *Manage Properties*.



This will open up a new menu used for setting up new properties and saving them for reuse in different documents. More on that later.

Property Styles

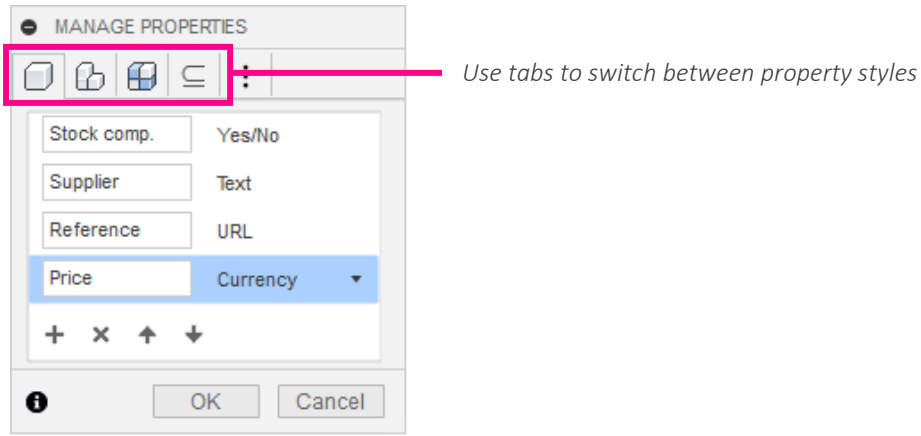
Properties are set up as part of a group, we call these groups the *Property Styles*. When changing the properties of one of your components you will have to choose which style they are part of; this will result in showing different properties. The following styles are available:

- Parts style
- Assembly style
- Design style

There is also an option available to add the properties to a common group. All properties added to this group will show in any of the previously described styles.

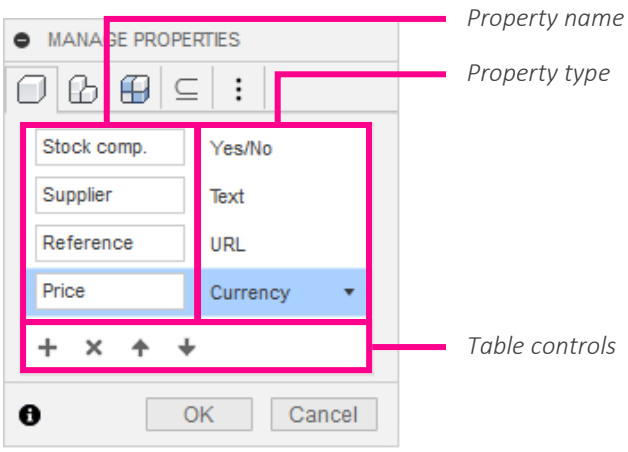
- Common properties

Note that you can switch between styles using the tab-menu at the top of the menu.



Adding properties

Choose any of the previously described styles by switching the active tab. The menu below will show a small table. Each row represents a custom property. When there are no properties set up yet this table will be empty. Click on the plus-icon at the bottom row of the table to add a new property.
The table exists of two columns, the first column will show the property name and the second column will show the property type. Fill in both. Note that once used the type cannot be changed since this might conflict with existing data.



The property types can be any of the following:

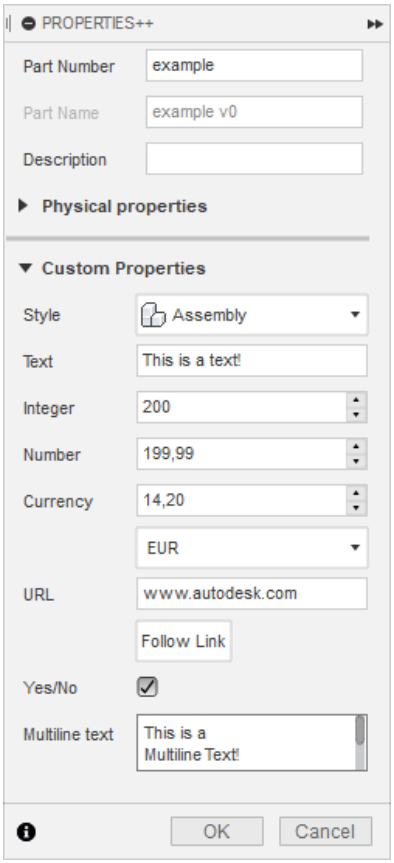
Text: *Normal single line text. This can contain anything from numbers to special characters. The exception is the ~ character which cannot be used.*

Integer: *These are real numbers (they do not have decimal places) which can also be zero or have a have negative value. For example, the value could be 50, 0 or -50. Integers should be between -2 million and 2 million.*

Number: *A number can contain any kind of value, including numbers with decimal places. For example, the value could be 50.00, 0.00 or -50.50. Numbers should be between -2 million and 2 million. Note that the amount of decimal places that are available will depend on your Fusion 360 preferences.*

Currency: *This will contain a Number (and all its properties) plus a dropdown-menu for the desired currency. Note that you can change the currency for each component. The standard currency (the currency a component will start with) can be set at the options tab in the property manager. See below.*

URL: *A URL is a reference to a file online or on your pc. Fill in the full path of the file in the shown textbox. A button will appear to open the file or path. For example; you will be able to reference a pdf file just like you will be able to reference a webpage.*



Yes/No: *The yes/no parameter will show a checkbox.*

Multiline text: *The multiline text can contain the same information as Text but will create a scrollable textbox for larger text inputs. The textbox is two lines high.*

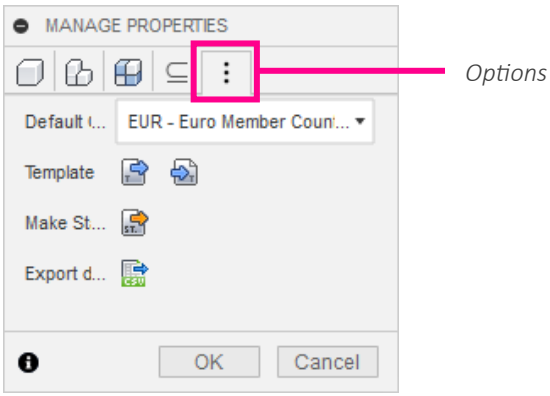
Table controls

The bottom row of the table will show you a couple of table controls. Using the + you can add more properties to the list. If you select a property, by clicking on the name or property type, you will be able to delete that property using ×.

The order of the properties in de component's Property++ menu will depend on the order in this menu. To change the order of the items use the ↓ or ↑ buttons after selecting the row to move.

Save and transfer property styles

If you would like to reuse your properties styles; go to the Options tab of the Properties Manager. Reusing styles will make sure that you do not have to setup your properties for every design. It will also help to improve the bill of material if you would use linked document (more on that later).



In the options tab you will find 2 methods of saving the current styles: You could save the styles as a *Template* and you could save them as a *Standard*.

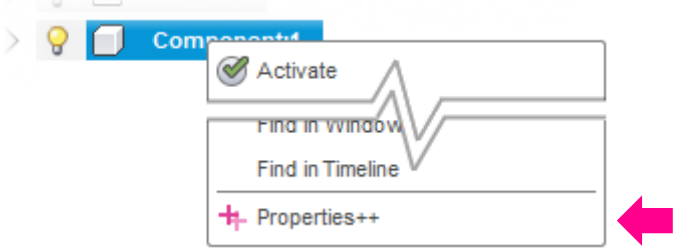
A Template will create an external file that you could place anywhere on your pc or network. Select the *Export Template* button and than select the location and name that you want to save your template to. This document can be reused by selecting the *Import Template* button. (Note the arrow of the icon pointing toward the document or away from the document.) Importing a template will add all the saved properties to your current set of properties or replace them alltogether.

The second methond of saving your properties is the *Standard*. When you save the styles as a standard, Properties++ will create a template file that automatically loads every time you start Properties++ with a new file. This will help you get the same properties in all your documents without the need of repetetive loading of files. The standard file is saved locally and can not be placed at a different location.
Note that the *Standard* option is not available in the free version of Properties++.

Pro tip: Internally, properties exist out of 3 components: their names, their type and their unique identifiers. When creating a Bill of Materials Properties++ will look for all properties that use the unique identifiers that are found in the setup styles. Reusing styles by Template or a Standard you will make sure it will be possible for Properties++ to also recognize properties in linked components as they will use the same unique identifiers. Also note that manually creating the same property in a different document will not result in the correct recognition since the unique identifier will be different.

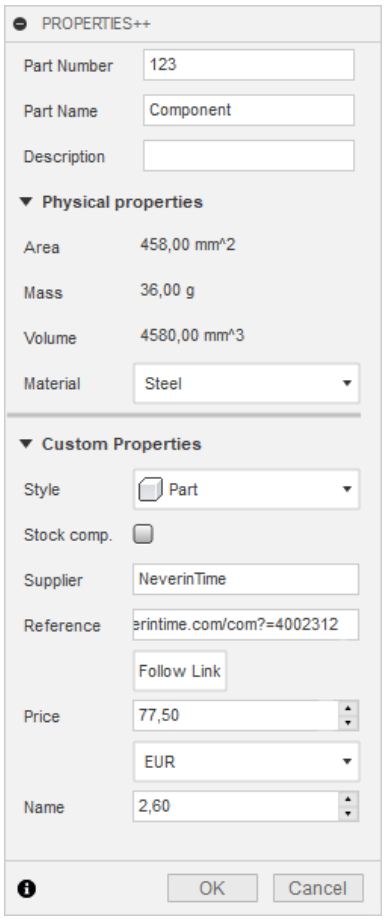
2. Add data per component

Once all properties are set up you can start using them. To add data to your component select a component in the (design) browser, right click and select Properties++.



This will open up the main Properties++ menu. The menu looks similar to Fusion 360's existing property menu with some small changes to the Physical Properties sub-menu and the addition of the custom Properties.

The physical properties show data on the design analysis like usual. Note that these results update every time you reopen the Properties++ menu. The shown units depend on your set preferences in Fusion 360.
The Material property is added to the existing list of physical properties. This dropdown menu will make it possible to quickly switch materials. The list exists out of your Favorite materials (as set with Fusion Manage Materials menu), or when you do not have any; all materials. You can only alter a components material if it exists out of a single body since multiple body's might have multiple materials.



Below you will find the Custom Properties. The list of custom properties always start with the Style options. As noted before, your custom properties are part of one of three styles, changing the style option changes which properties will be visible. Note that you can safely switch styles without losing previously filed data.

Below the style option you will find the properties that you have set up. The behavior of these options depend on the property type. You can read more on that in chapter 1.

Once you click on OK your new data will be saved. Only data that is visible through the correct property style will be saved though. Clicking on Cancel will cancel any changes and nothing is saved.

If you would like to change the properties of a linked component you will have to open the document separately. Although you can read it's properties from the assembly file, you cannot alter its properties.

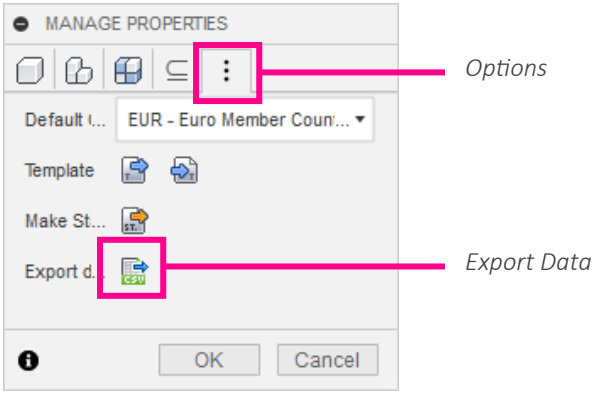
You can copy a component to a different design and still keep its properties. Make sure that the new design does have the same properties set up as the original. This cannot simply be done by making new properties with the same name, you will have to import properties using a template or a Standard. See more on how to transfer property styles in chapter 1.

3. Export to a table

A design's data can be exported to a table. This table will be saved in the CSV format (*comma separated value*) which is a format that can be opened by all major spreadsheet editing tools.

CSV files are basically text files, instead of using rows and columns it uses lines and delimiters. Due to the simplicity of this file format it became a universal format for many data tables. The delimiter that this file format uses is by standard the comma (,). However, on some computers the comma is used as a decimal sign, in those cases Properties++ will use the semicolon (;)as the delimiter

To export the data of your model we'll have to open Manage Properties menu again. Open the *MODIFY* panel and select Manage Properties. (The Modify panel is available in de Model workspace) Move to the last tab in this menu, here you will find the *Options*. Now select the button after Export Data.



Tip: Some spreadsheet editing tools, such as older versions of Microsoft Excel, do not automatically open CSV files using the correct settings. If this happens, open your spreadsheet software and then use the Open function to open the CSV file. This will give you some addition options to open the document correctly.

Properties++ will save the datasheet in a style similar to a Bill Of Materials. Although with this data all components will be separated in parts, assemblies and designs. The data in this table will correspond with the (saved) properties you have added to these styles in the Properties Manager.
There will be some additional columns in the datasheet: the quantities of a component within the main document (counting from every assembly), the component name, the component number, the component description, the component material and a parent parameter which will tell you how many subcomponents are found in this component (if any). The datasheet will show you all components of this model, independent if this is a subcomponent or not.

If you use linked files in your design you will need to make sure that these files use the same properties as the model you export the data from. If they do not contain the same properties they won't export their custom properties. (Note that creating properties with the same name and data type does not make it the same property) To prevent this you can copy your property styles using a *Template* or a *Standard*. See chapter 1 on how to use these.